

Various algorithms:

1. Linear Search
2. Binary Search
3. Jump Search
4. Interpolation Search
5. Exponential Search
6. Fibonacci Search
7. Bubble Sort
8. Selection Sort
9. Insertion Sort
10. Merge Sort
11. Quick Sort
12. Heap Sort
13. Timsort
14. IntroSort
15. Shell Sort
16. Counting Sort
17. Radix Sort
18. Bucket Sort
19. DepthFirst Search (DFS)
20. BreadthFirst Search (BFS)
21. Dijkstra's Algorithm
22. Topological Sorting
23. Kruskal's Algorithm
24. Prim's Algorithm
25. BellmanFord Algorithm
26. FloydWarshall Algorithm
27. Johnson's Algorithm
28. Tarjan's Strongly Connected Components
29. Kosaraju's Algorithm
30. EdmondsKarp Algorithm
31. Dynamic Programming (DP) Algorithms
32. Fibonacci Sequence (DP)
33. Knapsack Problem (0/1)
34. Longest Common Subsequence (LCS)
35. Longest Increasing Subsequence (LIS)
36. Matrix Chain Multiplication
37. Traveling Salesman Problem (TSP) with DP
38. Egg Dropping Puzzle
39. Palindrome Partitioning
40. Optimal Binary Search Tree (OBST)
41. String Matching (Naive)

42. KMP (KnuthMorrisPratt) Algorithm
43. RabinKarp Algorithm
44. Suffix Array
45. Suffix Tree
46. Z Algorithm
47. AhoCorasick Algorithm
48. Binary Tree Traversal (Inorder, Preorder, Postorder)
49. Binary Search Tree (BST) Operations
50. AVL Tree Balancing
51. RedBlack Tree Operations
52. Splay Tree Operations
53. Treap (Tree + Heap)
54. Segment Tree
55. Fenwick Tree (Binary Indexed Tree)
56. Lowest Common Ancestor (LCA)
57. Activity Selection Problem
58. Fractional Knapsack
59. Huffman Coding
60. Job Sequencing Problem
61. Minimum Spanning Tree (Kruskal or Prim)
62. Dijkstra's Algorithm
63. Divide and Conquer Algorithms
64. Merge Sort
65. Binary Search
66. Quick Sort
67. Closest Pair of Points
68. Strassen's Matrix Multiplication
69. Karatsuba Algorithm
70. Backtracking Algorithms
71. NQueens Problem
72. Subset Sum Problem
73. Sudoku Solver
74. Hamiltonian Cycle
75. Graph Coloring
76. Knight's Tour Problem
77. Greatest Common Divisor (GCD) Euclidean Algorithm
78. Sieve of Eratosthenes
79. Fast Fourier Transform (FFT)
80. Modular Exponentiation
81. MillerRabin Primality Test
82. Pollard's Rho Algorithm
83. Chinese Remainder Theorem
84. Randomized Quick Sort
85. Monte Carlo Algorithm

86. Las Vegas Algorithm
87. Karger's Algorithm
88. Reservoir Sampling
89. KMeans Clustering
90. Linear Regression
91. Decision Trees
92. Support Vector Machines (SVM)
93. Gradient Descent Optimization
94. Backpropagation
95. Reinforcement Learning (QLearning)
96. Line Segment Intersection
97. Convex Hull (Graham's Scan)
98. Closest Pair of Points
99. Sweep Line Algorithm
100. Voronoi Diagram
101. Huffman Coding
102. RSA Algorithm
103. AES (Advanced Encryption Standard)
104. LempelZivWelch (LZW) Compression
105. Elliptic Curve Cryptography (ECC)
106. Tower of Hanoi
107. BoyerMoore Majority Vote Algorithm
108. Reservoir Sampling
109. Hungarian Algorithm
110. Simplex Algorithm
111. Bloom Filter
112. Ternary Search
113. Binary Search Tree Search
114. DepthLimited Search
115. Iterative Deepening DFS
116. BestFirst Search
117. Comb Sort
118. Cocktail Shaker Sort
119. Gnome Sort
120. OddEven Sort
121. Patience Sort
122. Strand Sort
123. Cycle Sort
124. Pancake Sort
125. Bogo Sort
126. FordFulkerson Algorithm
127. Dinic's Algorithm
128. PushRelabel Algorithm
129. Bipartite Matching

130. Hungarian Algorithm (Graph Context)
131. Blossom Algorithm (Edmonds' Matching)
132. HopcroftKarp Algorithm
133. Borůvka's Algorithm
134. ChuLiu/Edmonds' Algorithm (Directed MST)
135. Articulation Points (Tarjan's)
136. Bridge Finding (Tarjan's)
137. Eulerian Path/Circuit
138. Fleury's Algorithm
139. Hierholzer's Algorithm
140. Longest Palindromic Subsequence
141. Longest Palindromic Substring
142. Maximum Subarray Sum (Kadane's Algorithm)
143. Weighted Job Scheduling
144. Cutting Rod Problem
145. Coin Change Problem
146. Minimum Cost Path
147. Word Break Problem
148. Box Stacking Problem
149. Longest Bitonic Subsequence
150. Manacher's Algorithm
151. BoyerMoore Algorithm
152. Horspool Algorithm
153. Sunday Algorithm
154. Finite Automaton Matcher
155. Suffix Automaton
156. BurrowsWheeler Transform
157. Ukkonen's Algorithm
158. BTree Operations
159. B+ Tree Operations
160. Trie Operations
161. Binary Indexed Tree (Fenwick Tree, repeated for context)
162. Cartesian Tree
163. LinkCut Tree
164. HeavyLight Decomposition
165. Centroid Decomposition
166. Range Minimum Query (RMQ)
167. Persistent Segment Tree
168. Interval Scheduling
169. Minimum Coin Change
170. Egyptian Fraction
171. Set Cover Approximation
172. Vertex Cover Approximation
173. Fast Inverse Square Root

174. Median of Medians
175. CooleyTukey FFT
176. SchonhageStrassen Multiplication
177. Rat in a Maze
178. Cryptarithmic Puzzle Solver
179. Tug of War Problem
180. MColoring Problem
181. Partition Problem
182. Extended Euclidean Algorithm
183. Euler's Totient Function
184. Lucas Theorem
185. Fast Modular Exponentiation
186. Shanks' BabyStep GiantStep
187. TonelliShanks Algorithm
188. BerlekampMassey Algorithm
189. Pollard's p1 Algorithm
190. Quadratic Sieve
191. General Number Field Sieve
192. Skip List Operations
193. Randomized Primality Testing (Fermat)
194. Randomized Binary Search
195. Treap (Randomized BST, repeated for context)
196. HyperLogLog (Cardinality Estimation)
197. CountMin Sketch
198. Naive Bayes Classifier
199. KNearest Neighbors (KNN)
200. Perceptron
201. AdaBoost
202. Gradient Boosting
203. Random Forest
204. ExpectationMaximization (EM)
205. Hidden Markov Model (HMM)
206. Monte Carlo Tree Search (MCTS)
207. AlphaBeta Pruning
208. Delaunay Triangulation
209. Fortune's Algorithm (Voronoi Diagram)
210. BentleyOttmann Algorithm
211. Point Location in Planar Subdivision
212. Rotating Calipers
213. Kirkpatrick's Point Location
214. RunLength Encoding (RLE)
215. Arithmetic Coding
216. ShannonFano Coding
217. DES (Data Encryption Standard)

218. Triple DES (3DES)
219. DiffieHellman Key Exchange
220. ElGamal Encryption
221. SHA (Secure Hash Algorithm) Family
222. MD5 (Message Digest 5)
223. FisherYates Shuffle
224. Disjoint Set Union (UnionFind)
225. Rolling Hash
226. LZ77 Compression
227. LZ78 Compression
228. GaleShapley Algorithm (Stable Matching)
229. Zobrist Hashing
230. Dancing Links (Algorithm X)
231. Simulated Annealing
232. Genetic Algorithm
233. Ant Colony Optimization
234. Particle Swarm Optimization
235. Cuckoo Search
236. Tabu Search
237. Hill Climbing
238. van Emde Boas Tree Search
239. Splay Tree Search
240. Fusion Tree Search
241. YFast Trie Search
242. Cuckoo Hashing Lookup
243. Flash Sort
244. Library Sort (Gapped Insertion Sort)
245. Smoothsort
246. Cartesian Tree Sort
247. Tournament Sort
248. Block Sort
249. Spreadsort
250. Mincut (StoerWagner Algorithm)
251. GomoryHu Tree
252. Klein's Algorithm (Planar Shortest Paths)
253. Suurballe's Algorithm (Disjoint Paths)
254. Christofides Algorithm (Approximate TSP)
255. HeldKarp Algorithm (Exact TSP)
256. Gabow's Algorithm (Minimum Degree Spanning Tree)
257. Chvátal's Algorithm (Set Cover on Graphs)
258. Thorup's Algorithm (SingleSource Shortest Paths)
259. Hagerup's Algorithm (AllPairs Shortest Paths)
260. KargerStein Algorithm (MinCut, Faster)
261. Maximum Cardinality Matching (MicaliVazirani)

262. Planar Separator Theorem (LiptonTarjan)
263. FruchtermanReingold (Graph Drawing)
264. Four Russians Algorithm (Boolean Matrix Multiplication)
265. Hirschberg's Algorithm (SpaceEfficient LCS)
266. Knuth's Optimization (DP Speedup)
267. DivideandConquer Optimization (DP Speedup)
268. Convex Hull Trick (DP Optimization)
269. LiChao Tree (DP Optimization)
270. Parallel Dynamic Programming (for Multicores)
271. Sparse Table (RMQ in $O(1)$)
272. Longest Common Substring (with Suffix Tree)
273. Monge Array Optimization
274. CrochemorePerrin Algorithm (TwoWay String Matching)
275. Suffix Array (DC3/Skew Algorithm)
276. LempelZiv Factorization
277. Lyndon Factorization
278. RunLength Encoded BurrowsWheeler Transform
279. FMIndex (FullText Search)
280. Longest Common Prefix (LCP) Array
281. Compressed Suffix Array
282. Weiner's Algorithm (Suffix Tree Construction)
283. McCreight's Algorithm (Suffix Tree Construction)
284. OrderStatistic Tree
285. Scapegoat Tree
286. AA Tree
287. Wavelet Tree
288. Range Tree
289. KD Tree (Multidimensional Search)
290. Quadtree
291. RTree (Spatial Indexing)
292. Dynamic Segment Tree (Lazy Propagation)
293. Persistent Binary Search Tree
294. Euler Tour Technique (Tree Flattening)
295. Tree Contraction (Parallel Tree Algorithms)
296. Greedy Coloring (Approximation)
297. Matroid Intersection Algorithm
298. Online Greedy Scheduling
299. Maximum Coverage Problem (Greedy Approximation)
300. Minimum Dominating Set (Greedy Approximation)
301. Fast Multipole Method (FMM)
302. Closest Pair of Points (Randomized)
303. ToomCook Multiplication
304. CoppersmithWinograd Matrix Multiplication
305. Fast WalshHadamard Transform

306. Densest Submatrix (Divide and Conquer)
307. Exact Cover Problem (Algorithm X)
308. Dynamic Backtracking (Constraint Satisfaction)
309. Branch and Bound (for Optimization Problems)
310. Dancing Links (DLX)
311. Combinatorial Generation (Gray Code)
312. AKS Primality Test
313. Dixon's Factorization
314. LenstraLenstraLovász (LLL) Algorithm
315. Continued Fraction Factorization
316. Number Field Sieve (NFS)
317. Discrete Logarithm (Pollard's Kangaroo)
318. Gröbner Basis Computation
319. Buchberger's Algorithm
320. Faugère's F4/F5 Algorithm
321. Integer Relation Detection (PSLQ)
322. Fast Matrix Exponentiation
323. Möbius Inversion Formula (Applications)
324. Frievalds' Algorithm (Matrix Multiplication Verification)
325. MinHash (Similarity Estimation)
326. CountSketch (Streaming Data)
327. Randomized Shell Sort
328. Random Walk Algorithms (Graph Sampling)
329. MCMC (Markov Chain Monte Carlo)
330. Gibbs Sampling
331. MetropolisHastings Algorithm
332. Randomized Rounding (Approximation)
333. Luby's Algorithm (Maximal Independent Set)
334. Deep QLearning (DQN)
335. Proximal Policy Optimization (PPO)
336. Generative Adversarial Networks (GANs) Training
337. Variational Autoencoders (VAEs)
338. Transformer Models (Attention Mechanism)
339. BERT Training (Bidirectional Encoder)
340. AlphaGo's MCTS (Monte Carlo Tree Search)
341. Policy Gradient Methods
342. Trust Region Policy Optimization (TRPO)
343. tSNE (Dimensionality Reduction)
344. UMAP (Uniform Manifold Approximation)
345. Chazelle's Triangulation (Polygon Triangulation)
346. Chan's Algorithm (Convex Hull)
347. ShamosHoey Algorithm (Line Segment Intersection)
348. Overmarsvan Leeuwen (Dynamic Convex Hull)
349. PreparataHong (DivideandConquer Convex Hull)

350. Randomized Incremental Construction (Delaunay)
351. Kinetic Data Structures (Moving Points)
352. Slab Method (Geometric Searching)
353. Trapezoidal Decomposition
354. BWT with MoveToFront (MTF)
355. ContextAdaptive Binary Arithmetic Coding (CABAC)
356. PAQ Compression
357. Homomorphic Encryption (e.g., Gentry's Scheme)
358. LatticeBased Cryptography (e.g., NTRU)
359. PostQuantum Cryptography (e.g., McEliece)
360. ZeroKnowledge Proofs (zkSNARKs)
361. Secure Multiparty Computation (Yao's Protocol)
362. Shamir's Secret Sharing
363. BLS Signature (BonehLynnShacham)
364. BakerCampbellHausdorff Formula (Lie Groups)
365. Coppersmith's Algorithm (LowExponent RSA Attack)
366. Berlekamp's Algorithm (Polynomial Factorization)
367. Risch Algorithm (Symbolic Integration)
368. KnuthBendix Completion Algorithm
369. DPLL Algorithm (SAT Solving)
370. ConflictDriven Clause Learning (CDCL)
371. Chaff Algorithm (SAT Solver)
372. WalkSAT (Local Search for SAT)
373. GaussNewton Algorithm (Nonlinear Least Squares)
374. LevenbergMarquardt Algorithm
375. Karmarkar's Algorithm (Interior Point Method)
376. ellipsoid Method (Linear Programming)
377. Cutting Plane Method (Integer Programming)
378. Benders Decomposition
379. Vogel's Approximation Method (Transportation Problem)
380. Hungarian Method (Weighted Bipartite Matching)
381. PushRelabel (Max Flow, Repeated for Context)
382. Sweep and Prune (Collision Detection)
383. BarnesHut Algorithm (NBody Simulation)
384. Quantum Fourier Transform (Shor's Algorithm)
385. Grover's Algorithm (Quantum Search)
386. HHL Algorithm (Quantum Linear Systems)
387. DeutschJozsa Algorithm (Quantum)
388. LRU Page Replacement Algorithm
389. Banker's Algorithm with Resource Request Handling
390. EDF (Earliest Deadline First) with Dynamic Priority Adjustment
391. TwoPhase Commit with Recovery (2PCR)
392. LogStructured MergeTree (LSMTree) Compaction
393. NUMAAware Scheduling (NonUniform Memory Access)

394. CFS (Completely Fair Scheduler) with Bandwidth Control
395. Kyber (LatticeBased Key Encapsulation Mechanism)
396. Dilithium (LatticeBased Digital Signature)
397. SPHINCS+ (Stateless HashBased Signature)
398. ZeroKnowledge Proofs (zkSNARKs)
399. Honey Encryption with DistributionTransforming Encoder (DTE)
400. Moving Target Defense (MTD) with GameTheoretic Optimization
401. Adversarial Machine Learning Defense (Robust Optimization)
402. Variational Quantum Eigensolver (VQE) for Neural Networks
403. AlphaCode's TransformerBased Code Generation
404. Grok's DeepSearch Iterative Web Analysis (Hypothetical)
405. Contrastive LanguageImage Pretraining (CLIP) with Adversarial FineTuning
406. Neural Architecture Search (NAS) with Reinforcement Learning
407. Deep Deterministic Policy Gradient (DDPG) with Hindsight ExperienceReplay
408. Graph Neural Network (GNN) with Attention for Dynamic Graphs
409. Complex Algorithms in Quantum Computing
410. Shor's Algorithm with Error Correction
411. Grover's Algorithm with Amplitude Amplification
412. HHL Algorithm (HarrowHassidimLloyd)
413. Quantum Approximate Optimization Algorithm (QAOA) with MultiLayer Ansatz
414. BiasField Digitized Counterdiabatic Quantum Optimization (BFDCQO)
415. Topological Quantum Error Correction (Surface Code)
416. Quantum Fourier Transform (QFT) with Fault Tolerance
417. Quantum Phase Estimation (QPE) with Iterative Refinement
418. Complex Algorithms in Other Leading Domains
419. Paxos with Failure Detectors (ChandraToueg)
420. Raft with Log Compaction and Snapshotting
421. Byzantine Fault Tolerance (BFT) with Honey Badger Optimizations
422. De Bruijn Graph Assembly with Error Correction
423. NeedlemanWunsch with Affine Gap Penalties
424. Hidden Markov Model (HMM) with Viterbi for Gene Prediction
425. Proof of Stake (PoS) with Ouroboros Praos
426. Shamir's Secret Sharing with Threshold Cryptography
427. BlackScholes with Monte Carlo for Option Pricing
428. Heston Model with Fourier Transform Pricing
429. LongstaffSchwartz Algorithm for American Options
430. Wavelet Transform with Daubechies Filters
431. Kalman Filter with Unscented Transform (UKF)
432. Compressive Sensing with Basis Pursuit Denoising